

JAGAN SHANMUGAM

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Experience

MBiton GmbH (Mercedes-Benz Group)

Jun 2020 – Present

Expert Data Scientist

Berlin, Germany

System Copilot - Agentic Workflows for MBSE | *LangGraph, GraphQL, Bedrock AgentCore, AWS*

- Architected Langgraph-based agent backend with memory, observability for Systems & Function Model (SFM) data in Model Based Systems Engineering (MBSE) platform.
- Developed an DSPy-based leave-one-out eval harness for artifact creation that measures F1/Recall/Precision, hallucination rate and tool selection on live GraphQL backend.

House of AI - Internal AI Platform | *LangGraph, LlamaIndex, PGVector, Azure OpenAI, Langfuse, DSPy, AI SDK*

- Built and shipped "House of AI" - a platform serving 100+ teams across use cases including multi-persona agentic chat, custom knowledge assistants for scattered docs, debugging build pipelines by log analysis.
- Designed and implemented multi-source agentic RAG backend with hybrid search and query fusion for higher recall with LangGraph-based agent runtime with tool calling, stateful orchestrations and safeguards for long context workflows.
- Built robust data ingestion pipelines supporting multimodal content vectorization from enterprise sources (Confluence, JIRA, GitLab, IBM DNG) and developed a data parser microservice with celery utilizing Docling and MarkItDown.
- Built an offline evaluation harness with synthetic dataset generation and LLM-as-a-Judge scoring for continual learning.
- Established production monitoring framework tracking helpfulness, response quality, relevance of sources, and user satisfaction metrics that drove continuous agent improvements through system prompt optimization.

ExploreMe - HeyMercedes Voice Assistant | *BERT, MLFlow, Databricks, PromptFlow, Azure CognitiveSearch*

- Finetuned a multilingual BERT-based binary classifier to identify information requests from all utterances on a curated dataset for six languages and evaluated it for several iterations with expert annotations reaching upto 98% recall.
- Integrated the classifier as a micro-service FastAPI backend via Kubernetes serving requests from drivers on-road.
- Developed an unsupervised intent clustering and annotation prep pipeline using sentence transformer embeddings and UMAP, HDBSCAN and bayesian hyperparameter optimization to accelerate annotation process and intent discovery.
- With data-driven feature development and prioritization, identified 6 features that led to an increase in feature coverage by 51%. Continuously analyzed PROD utterances and benchmarked NLU providers to detect scope gaps, and derived improvements for existing voice assistant domains.

Telelog - Driver & Vehicle Data Analytics | *Tableau, Databricks, PySpark*

- Analysed 500K+ daily driving sessions across 16 countries and 8 car lines, processing 1M+ telemetry signals/day to quantify usage patterns across 6 HVAC features and infotainment controls for data-driven feature prioritization.

Scientific Games, India Pvt Limited

Oct 2014 – Aug 2017

Software Engineer - Business Intelligence

Bengaluru, India

- Built ETL pipelines (SSIS, SQL Server) for BI data integration for Gaming Corporations.

Education

University of Stuttgart

Grade: 1.5, Oct 2017 – Jan 2020

Master of Science in Information Technology - Focus on Machine/Deep Learning

Stuttgart, Baden-Württemberg

Master Thesis & Research Internship | *Bosch Center for Artificial Intelligence, Renningen | Apr 2019 – Dec 2019*

- Latent space Bayesian Optimization with Transfer Learning: 🏆 - Developed a method for warm starting Bayesian Optimization (BO) of the target task with previous optimization data from related tasks for finding optimal parameter settings in industrial processes.

Student Research Assistant | *Machine Learning and Robotics Lab, Stuttgart | Dec 2018 – Mar 2019*

Work Student | *Daimler AG, Sindelfingen | Mar 2018 – Jan 2019*

College of Engineering, Guindy

Grade: 85%, Aug 2010 – May 2014

Bachelor of Engineering in Electronics and Communication

Chennai, India

Technical Skills

ML/LLM: PyTorch, Pandas, Transformers, LLaMaIndex, DSPy, LangGraph, AI SDK, MCP

Data/Infra/DevOps: PySpark, Airflow, MLFlow, Langfuse, Databricks, Docker, Kubernetes, Terraform, Azure, AWS

Databases/Languages: Postgres, MS SQL Server, Python, TypeScript, SQL, FastAPI, React, NextJS

Leadership / Awards

- Mentoring junior data scientists and supervised bachelor and master thesis students in MBiton.
- Co-organizing Generative AI CoP (Community of Practice) and delivering talks on recent developments in the space.
- Won 2nd place in Bosch Agentic AI Hackathon on July 2025 developing Voice AI Agent platform based on LiveKit.
- Recipient of Deutschland Stipendium - winter semester 2019/20, for excellent academic performance in Masters.